Student Management System using MVC Architecture

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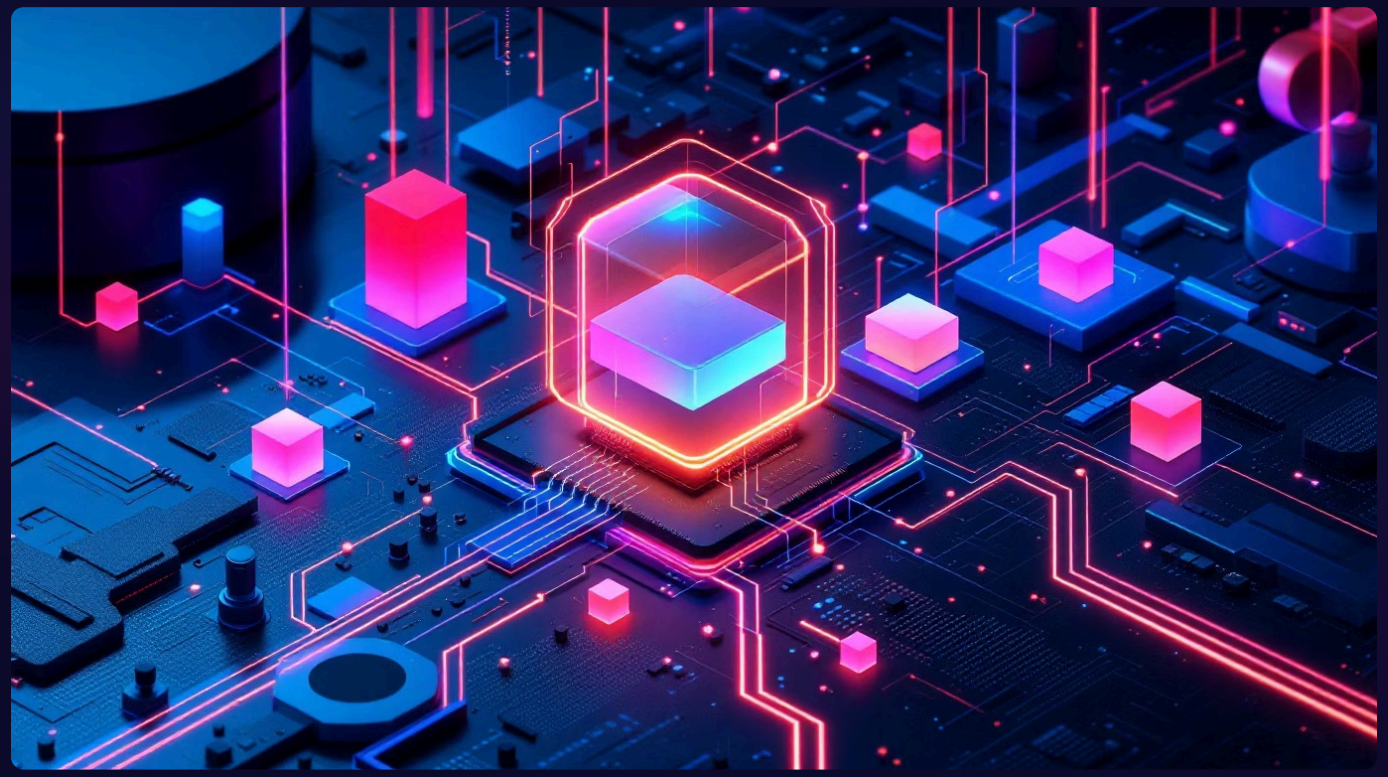
Introduction

Object-Oriented Programming (OOP) organises code into reusable, modular components using objects and classes. This project demonstrates OOP implementation in Java using MVC architecture.

The MVC pattern divides applications into three interconnected components, improving maintainability, scalability, and code readability.



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Problem Statement

Manual Inefficiency

Manual student information management is time-consuming and error-prone.

Data Organisation

Educational institutions need structured systems for managing student data efficiently.

Scalable Solution

Develop organised, scalable Student Management System using OOP and MVC architecture.

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Project Objectives

01

Implement OOP Concepts

Apply encapsulation, abstraction, inheritance, and polymorphism principles.

02

Design MVC Architecture

Create structured system with clear separation of concerns.

03

Manage Student Data

Efficiently handle student information including names, IDs, and records.

04

Build Modular System

Develop scalable and maintainable application structure.

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Technologies Used

Java

Core programming language for implementation

Object-Oriented Programming

Programming paradigm using objects and classes

MVC Architecture

Model-View-Controller design pattern

GitHub

Platform for version control and collaboration

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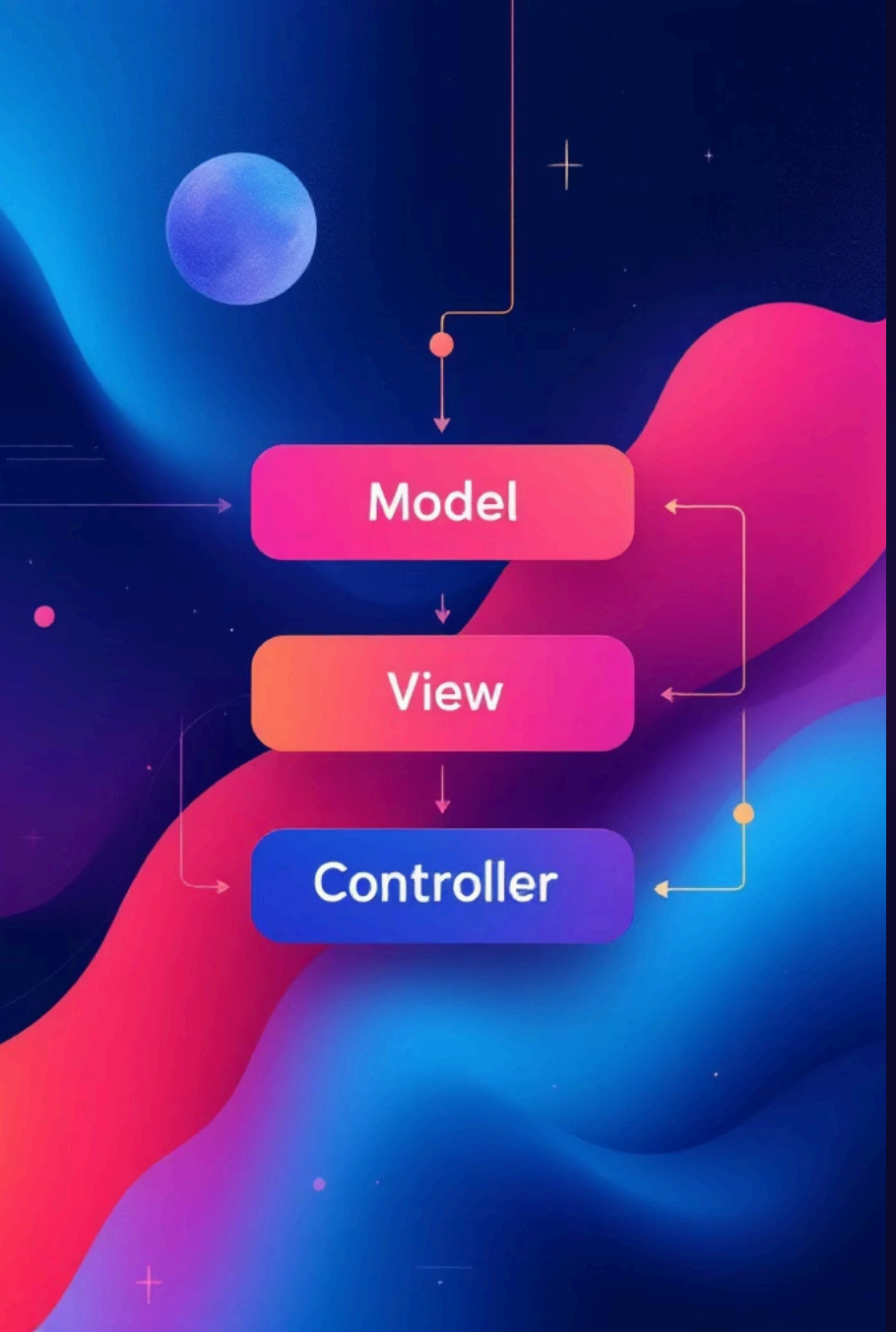

MVC Architecture

Model

Handles data and business logic through Student.java

View

Displays output to users via Studentview.java

Controller

Interfaces between Model and View using Studentcontroller.java

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OOP Concepts Implemented

Encapsulation

Binds data and methods together, accessing data through getter and setter methods.

Abstraction

Exposes only relevant details to users while hiding internal implementation complexity.

Inheritance

Enables code reuse and reduces redundancy, extendable for future enhancements.

Polymorphism

Allows methods to perform different tasks based on context and requirements.

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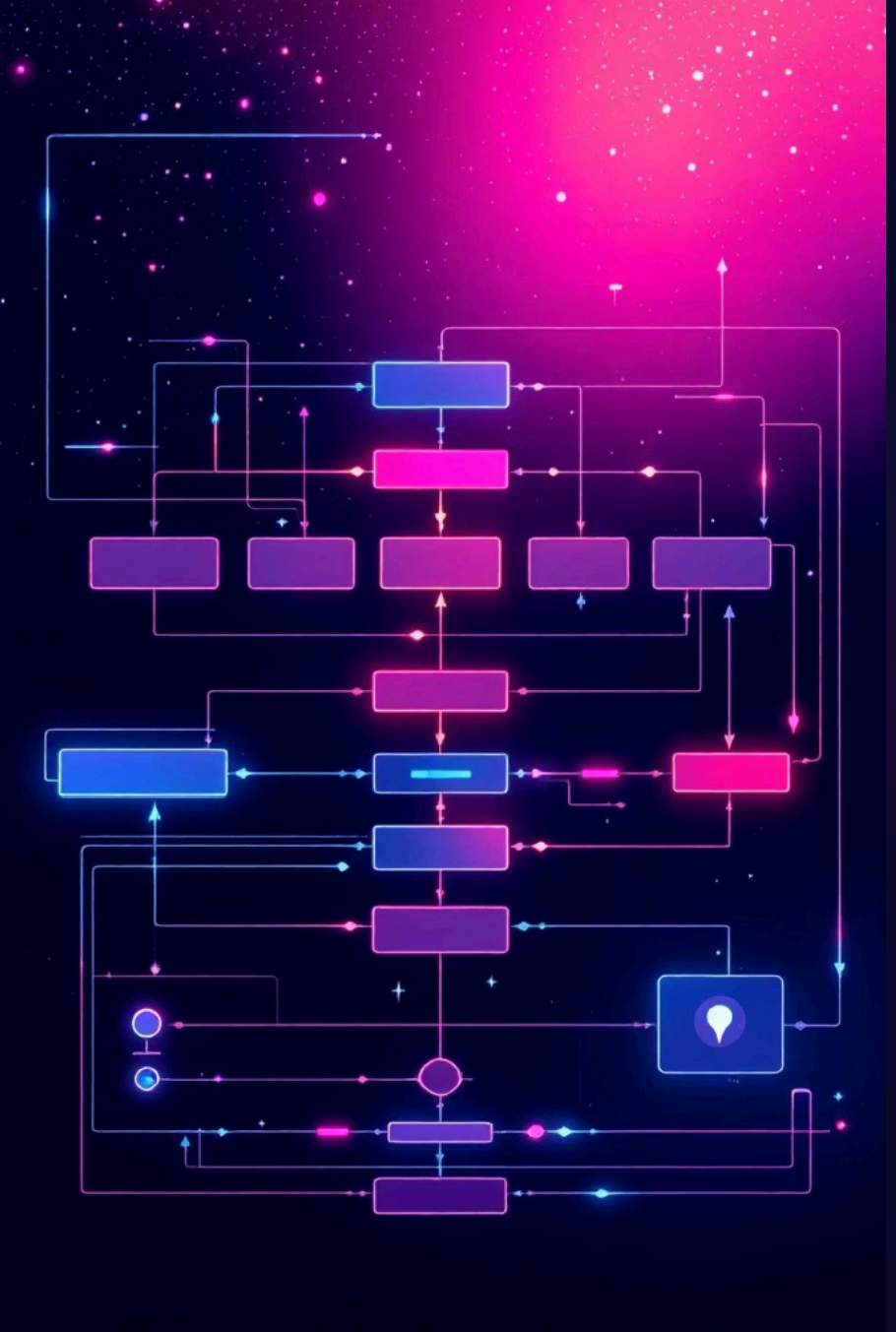

System Working Flow

User Interaction

User inputs student details through console interface

Controller Processing

Studentcontroller.java receives and processes user requests

Model Storage

Student.java processes and stores data securely

View Display

Studentview.java shows confirmation and output to user

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Advantages & Future Scope

Current Advantages

- Modular and structured design
- Easy maintenance and updates
- Reusable code components
- Clear separation of concerns

Future Enhancements

- Database integration with MySQL
- GUI application development
- Web-based system implementation
- Authentication and security features

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Conclusion

> This project successfully demonstrates Object-Oriented Programming concepts using MVC architecture, providing a structured approach to managing student data with enhanced maintainability and scalability.

The system establishes a solid foundation that can be enhanced through database integration, GUI development, and advanced security features for real-world educational applications.

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